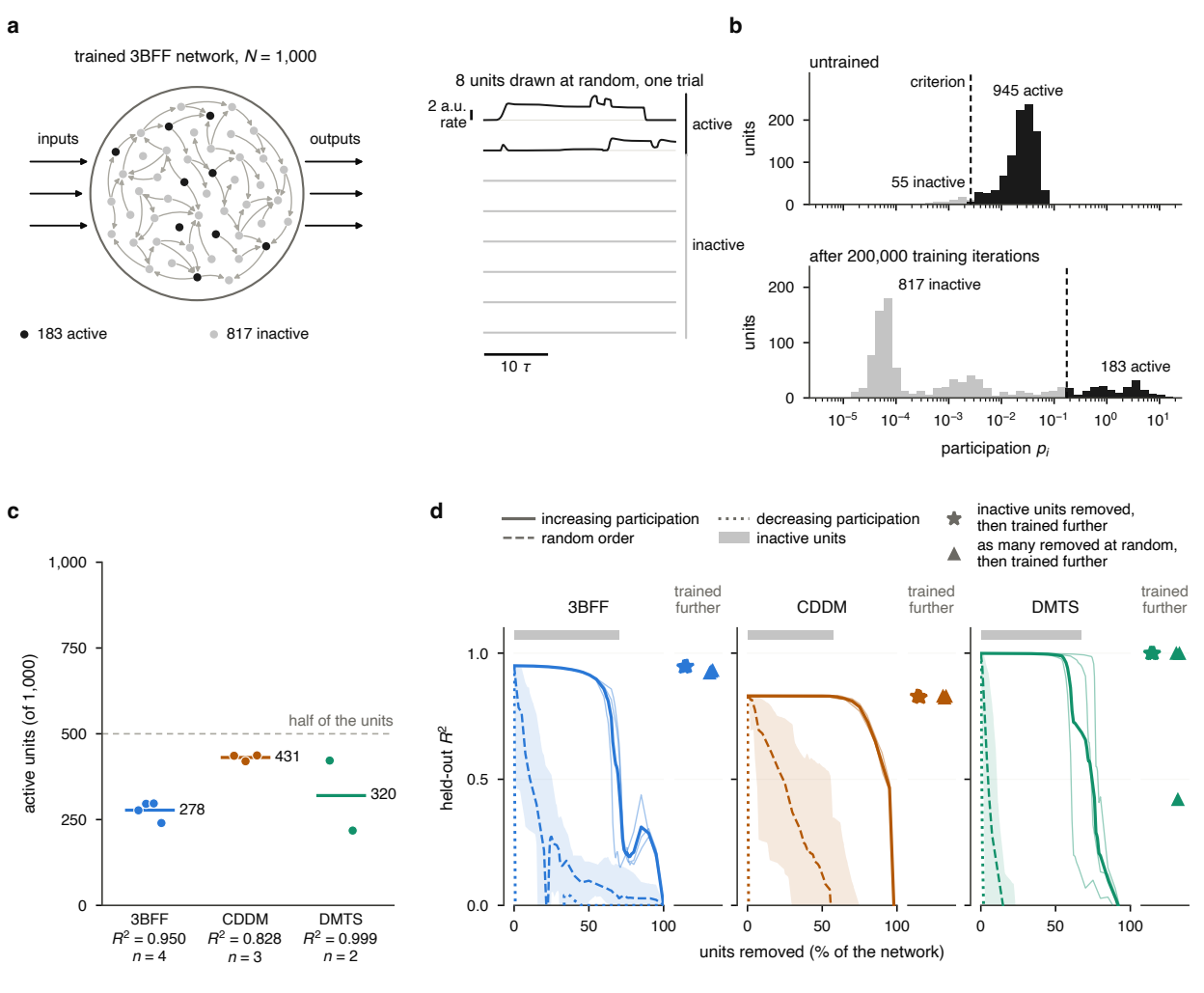
a
trained 3BFF network, N = 1,000
inputs
outputs
183 active
817 inactive
8 units drawn at random, one trial
2 a.u.
rate
active
inactive
10 τ
b
untrained
criterion
945 active
55 inactive
units
200
100
0
after 200,000 training iterations
817 inactive
183 active
10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1}
participation p_i
c
active units (of 1,000)
1,000
750
500
250
0
half of the units
431
278
320
3BFF
R^2 = 0.950
n = 4
CDDM
R^2 = 0.828
n = 3
DMTS
R^2 = 0.999
n = 2
d
increasing participation
random order
decreasing participation
inactive units
inactive units removed, then trained further
as many removed at random, then trained further
3BFF
CDDM
DMTS
trained further
held-out R^2
1.0
0.5
0.0
0
50
100
units removed (% of the network)